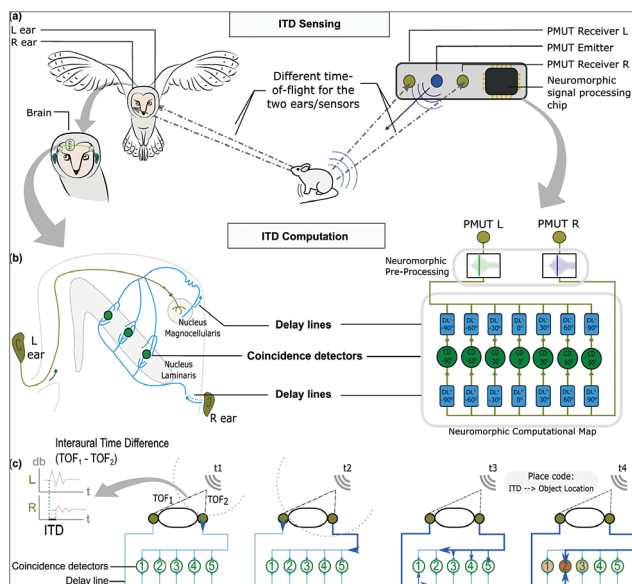


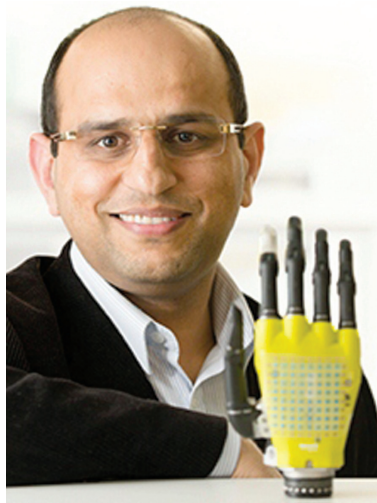


Object localisation system in barn owls and the proposed bio-inspired technology (Credit: www.nature.com)

ics this process. Mainly, it focuses on the acoustic based, object-localisation task. The setup emulates a particular type of neuron and delivers a spike-like pulse once specific time and amplitude dependencies have been met from the envelope detector. If the reflecting object is within earshot distance, the pipeline delivers a spike on a channel whose time difference is related to the target azimuth angle.

Biodegradable crop sensors and organic PVs being developed

Under the direction of Glasgow University researchers, green electronics for linked farming, such as biodegradable



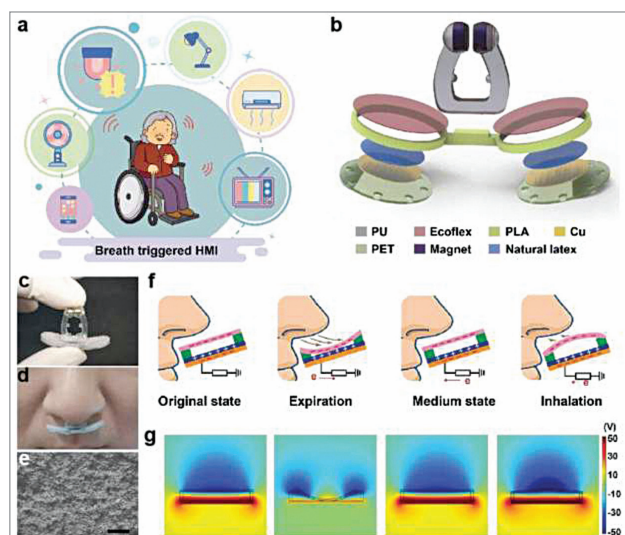
Prof. Ravinder Dahiya, James Watt School of Engineering, University of Glasgow with the prototype (Credit: University of Glasgow)

crop sensors and organic photovoltaics (PVs), are being developed. A modular sensor system for crop monitoring is being investigated and its two component pieces will either decompose into the soil at the end of their useful lives or can be recovered and utilised again in other systems. The biodegradable element will be the sensor patch itself. The sensors will be

able to perform tasks like monitoring pH, temperature, and bioimpedance. They will also have an electrical module with wireless communication capabilities. Reusability and repairability are top priorities in this module's design, which aims to reduce system waste as much as possible.

A device that fits into nostrils can manage smart devices by altering breathing patterns

A simple prototype gadget developed by Case Western Reserve University researchers allows users to manage 'smart home' devices by altering their breathing patterns.



Structural design and working mechanism of the breathing-driven triboelectric sensor (Credit: Advanced Materials Interfaces)

People with limited mobility or who have trouble speaking effectively may benefit from the self-powered device, which fits into the nostrils. Additionally, it can be set up to automatically notify medical staff when someone has a respiratory problem. It was created via a process called triboelectrification, and the tech is known as triboelectric nanogenerators (TENGs). TENGs can convert mechanical energy gathered from the environment into electricity for recharging consumer electronics or powering tiny devices like sensors.

A mathematical model that can solve next-generation battery issues

Stanford researchers have developed a new mathematical model that combines the physics and chemistry of very promising lithium-metal batteries. These batteries have a higher energy capacity, charge faster, and are lighter than